Benchmark Dirac–Debye resonances from a certified exact-structure spectral method: threshold-amplified spin-symmetry breaking of autoionization widths

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We compute resonances of the radial Dirac equation with the exponentially screened Coulomb (Yukawa–Debye) potential by two independent methods: a complex-scaled spectral discretization whose basis carries the exact Coulomb–Dirac cusp exponent, assembled from run-time–certified matrices and solved through a mass-shifted complex-symmetric pencil, and outgoing-wave (Gamow) shooting on the radial system matched to exact Riccati–Hankel free spinors. The two methods agree to ten–twelve digits on narrow and moderate resonances, and the spectral values are stable under rotation angle, basis size and scale to one part in 10^{10}. Against the only published Dirac–Yukawa resonance tables [Shi *et al.*, Few-Body Syst. **55**, 135 (2014)] we reproduce eight of twelve tabulated states at their printed precision and upgrade them to ten-digit benchmarks; for the narrow states at their strongest coupling our two methods concur that the published positions are off by $\sim 10^{-2}$ and the published widths are overestimated by up to two orders of magnitude, a pattern already visible internally in that work's nonrelativistic-limit comparison; and we resolve the widths of two near-bound states reported there as zero. Physically we map the fine-structure (spin-doublet) splitting of the $2p$ autoionization widths across the full coupling range. Near the critical screening the splitting is amplified by orders of magnitude: at fixed reduced screening $\lambda/Z = 0.24$ the ratio $\Gamma_{2p_{3/2}}/\Gamma_{2p_{1/2}}$ grows from 1.000 at $Z = 1$ to 77.5 at $Z = 80$, and for $Z \gtrsim 92$ the $2p_{1/2}$ partner remains *bound* while the $2p_{3/2}$ autoionizes — complete breaking of the near-degeneracy observed at weak coupling and high angular momentum. Deep in the continuum the same doublet barely splits, identifying the effect as a threshold phenomenon: the relativistic splitting of the critical screenings made dynamic. The resonances are born at the critical screenings of the companion bound-state map with the Wigner p-wave threshold law (measured exponent 1.530 vs. 3/2), and the position splitting reverses sign between the threshold and deep-continuum regimes.

I. INTRODUCTION

Resonances of screened Coulomb systems are a recurring theme across atomic physics, plasma physics and quantum chemistry: the Yukawa or Debye–Hückel potential models static screening in plasmas and electrolytes, and the fate of atomic levels as screening increases—their approach to threshold, dissolution into the continuum, and the widths of the resulting resonances—controls opacities, line shapes and ionization balance in dense environments [1, 2]. The standard computational tool is complex scaling [3–6], whose extension to the Dirac equation has a long history: the mathematical foundation was laid by Šeba [7] and by Balslev and Helffer [8], complex rotation was applied to the Dirac–Stark problem by Ivanov and Ho [9], a relativistic J-matrix formulation was given by Alhaidari [10], and the method is in production use in relativistic mean-field nuclear physics [11] and in complex-scaled relativistic configuration interaction [12].

For the Dirac–Yukawa problem itself the reference calculation is that of Shi, Liu, Niu and Guo [13], who applied complex scaling in a harmonic-oscillator basis, verified their nonrelativistic limit against the J-matrix benchmarks of Alhaidari, Bahlouli and Abdelmonem [14], and made a physical observation that frames the present work: the energies and widths of spin doublets — fine-structure partners $j = \ell \pm \frac{1}{2}$ — are approximately degenerate, preserving a good spin symmetry, in the regime they studied (high angular momenta $\ell = 5$ and 10). Spin- and pseudospin-symmetry properties of resonant states are an active topic in the nuclear-structure community [15].

This paper does two things. First, it establishes benchmark-grade Dirac–Yukawa resonances by a certified double method: a complex-scaled spectral discretization built on the exact-structure machinery of a companion paper [16] — a Laguerre basis carrying the analytic cusp exponent $\gamma = \sqrt{\kappa^2 - (Z\alpha)^2}$, exact generalized Gauss–Laguerre quadrature, run-time certification of the assembled operator identities, and a mass-shifted pencil that removes the rest-mass cancellation — cross-validated against an entirely independent outgoing-wave shooting calculation. Where the two methods overlap they agree to ten–twelve digits, which allows us both to extend the tables of Ref. [13] in precision and to adjudicate them: their narrow states at strong coupling are shifted and their widths substantially overestimated, an artifact consistent with harmonic-oscillator basis truncation near threshold and, internally, with the deviations their own nonrelativistic limit shows against the J-matrix values for precisely such states.

Second, and mainly, it maps what happens to the spin symmetry of autoionization when the coupling is strong and the states are near their thresholds — the atomic

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regime, opposite in both respects to that of Ref. [13]. The bound-state companion study [16] found that the critical screenings $\lambda_c(Z,\kappa)$ of fine-structure partners, degenerate nonrelativistically, split strongly with $Z\alpha$. Here we show the dynamical counterpart: near threshold the width ratio of the $2p$ doublet grows by orders of magnitude with $Z\alpha$, terminating in a regime where one partner is still bound while the other autoionizes, whereas deep in the continuum the same doublet remains nearly symmetric. The spin-symmetry breaking of resonance widths is thus a *threshold-amplified* effect, controlled by the relativistic splitting of the critical screenings.

Hartree atomic units ($\hbar = m = 1$, $c = 1/\alpha = 137.035\,999\,084$) are used throughout; the potential is $V(r) = -Ze^{-\lambda r}/r$ and $s = \lambda/Z$ denotes the reduced screening. Energies are binding-form, $E - c^2$, and resonances are written $E_r - i\Gamma/2$.

II. METHODS

A. Complex-scaled exact-structure discretization

Under the dilation $r \to re^{i\theta}$ every term of the radial Dirac coupling operators $\mathcal{O}_\pm = \mp d/dr + \kappa/r$ scales uniformly by $e^{-i\theta}$, the mass term is unaffected, and the Coulombic part of the potential scales exactly: the discretized, mass-shifted pencil is

$$H_\theta - c^2 S = \begin{pmatrix} e^{-i\theta}V_d & c\,e^{-i\theta}K \\ c\,e^{-i\theta}K^T & e^{-i\theta}V_d - 2c^2 S \end{pmatrix}, \qquad (1)$$

built from the *same real, run-time–certified matrices* K, V_d, S as the Hermitian solver of Ref. [16]; for the screened potential the smooth factor $e^{-\lambda r e^{i\theta}}$ enters the (converged) quadrature. The matrix is complex symmetric, and two structural facts carry over from the Hermitian theory exactly. First, the uniform $e^{-i\theta}$ scaling leaves the indicial equation at the origin invariant, so γ is unchanged and the cusp-adapted basis remains exact at $r \to 0$ under complex scaling. Second, the singular-value fiber decomposition of the free discretized operator [17] survives with $\sigma \to \sigma e^{-i\theta}$: the free spectrum of Eq. (1) lies *exactly* on the two rotated branches $\pm\sqrt{c^4 + c^2\sigma_j^2 e^{-2i\theta}} - c^2$, which we verify at the 10^{-10} level and which pins the rotated continua analytically. At $\theta = 0$ the solver reproduces the Hermitian one to 3×10^{-11}, and bound states are θ-independent to 1.4×10^{-10} over $\theta \in [0.05, 0.4]$ (discrete Balslev–Combes).

Resonances are identified by θ-*stability*: eigenvalues of the rotated-continuum strings move with θ while resonance eigenvalues do not, so the resonance is selected as the eigenvalue with the smallest displacement between two rotation angles. This selector proved essential: window-based selection can capture rotated-continuum artifacts that are static in other parameters. Every tabulated state is computed on a grid of two rotation angles, two basis sizes and two basis scales, and the maximum spread over the grid is quoted as its uncertainty; broad states ($\Gamma/E_r \gtrsim 2$) require the larger angle pair $\theta = (0.65, 0.80)$.

TABLE I. Double-method validation: $2p$ resonances $E_r - i\Gamma/2$ (a.u.) of $V = -Ze^{-\lambda r}/r$ at $s = \lambda/Z = 0.24$. "Spectral" is the θ-stable complex-scaled value; "shooting" the independent Gamow calculation.

Z, κ	spectral	shooting	rel. dev.
40, +1	2.941245536171	2.941245536305	4×10^{-11}
	$-1.355551909824\,i$	$-1.355551909813\,i$	
40, −2	3.413196281651	3.413196281388	7×10^{-11}
	$-1.806500986794\,i$	$-1.806500986677\,i$	
80, +1	0.848891759808	0.848891757494	3×10^{-9}
	$-0.079848876487\,i$	$-0.079848877035\,i$	
80, −2	12.673150466581	12.673150466566	1×10^{-12}
	$-6.188018766904\,i$	$-6.188018766901\,i$	

B. Independent validation: Gamow shooting

The second method shares no code or discretization with the first. The radial Dirac system is integrated outward from $r_0 = 10^{-8}$ with the exact indicial direction and a first-order series start, on a logarithmic grid to $r = 1$ (removing the $1/r$ stiffness) and in r beyond; at the matching radius the incoming-wave coefficient is extracted as the Wronskian of the regular solution with the *exact free outgoing spinor*, whose upper component is the Riccati–Hankel function $\hat{h}_\ell(pr)$, $p = \sqrt{E^2 - c^4}/c$, and whose lower component follows from the Dirac system itself. A complex secant iteration drives this coefficient to zero (the Siegert condition). The centrifugal term in the matching spinor is not optional: plane-wave matching leaves $O(\kappa/pR)$ errors that we measured at the 10^{-1} level before adopting the exact spinors. The method is limited to $\Gamma \lesssim 4$ a.u.: for broader states the outgoing component dominates the incoming one by more than the double-precision range at any usable matching radius.

Table I shows the agreement between the two methods: 4×10^{-11} and 7×10^{-11} relative for the $Z = 40$ doublet, 1.1×10^{-12} for the $2p_{3/2}$ resonance at $Z = 80$, with matching-radius stability at the 10^{-10} level. In the nonrelativistic limit ($Z = 1$) both channels reproduce the triple-verified benchmark of the companion nonrelativistic engine [16] with deviations of the $O(\alpha^2)$ magnitude and correct sign.

III. VALIDATION AGAINST AND ADJUDICATION OF THE PUBLISHED TABLES

Reference [13] tabulates twelve resonances (four printed digits) for spin doublets at $\ell = 5$ and $\ell = 10$ with strong couplings $A = 110$–200 ($A/c = 0.80$–1.46),

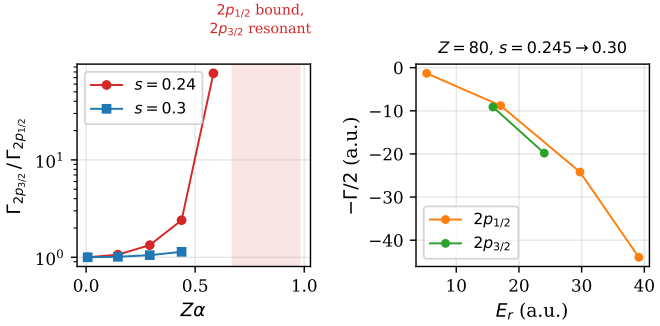


FIG. 1. Left: width ratio of the $2p$ fine-structure doublet vs. $Z\alpha$ at fixed reduced screening $s = \lambda/Z$. Near threshold ($s = 0.24$, red) the spin-symmetry breaking is amplified by orders of magnitude and terminates in the shaded region where the $2p_{1/2}$ is still bound while the $2p_{3/2}$ autoionizes; deep in the continuum ($s = 0.30$, blue) the same doublet stays nearly symmetric. Right: complex-plane trajectories of the doublet at $Z = 80$ as the screening sweeps $s = 0.245 \to 0.30$; the $2p_{3/2}$ dissolves early while the $2p_{1/2}$ traverses the fourth quadrant.

in the same units used here. Our double method reproduces eight of the twelve at their printed rounding — e.g. their $2.2834 - 0.0549i$ ($\kappa = 5$, $A = 110$) is our $2.283366856 - 0.054871772i$, with spectral–shooting agreement at 1.5×10^{-9} — thereby upgrading them to ten-digit benchmarks.

For the narrow states at $A = 170$, however, both of our methods concur on values incompatible with the published ones: their $0.4648 - 0.0066i$ ($\kappa = 5$) is $0.479288180 - 0.000051244i$ here (methods agreeing to 1.5×10^{-11}), and their $0.6613 - 0.0107i$ ($\kappa = -6$) is $0.669863069 - 0.000331590i$: positions shifted by $\sim 1.5 \times 10^{-2}$ and widths overestimated by up to two orders of magnitude. Two observations support the adjudication. First, the same pattern appears *within* Ref. [13]: for narrow states their nonrelativistic-limit values deviate from the J-matrix benchmarks they compare against (e.g. a width of 0.0168 against the J-matrix 1.09×10^{-6}), while for moderate and broad states the agreement is good. Second, the failure mode is the expected one for a harmonic-oscillator basis near threshold: narrow near-threshold states are spatially extended and carry the r^γ core behaviour, precisely the regime in which mismatched-basis convergence is slowest — quantified by the universal-exponent rate theorem of the companion analysis [17]. Finally, for the two near-bound states printed there as width zero we resolve $\Gamma = 3.04 \times 10^{-7}$ ($\kappa = 10$, $A = 200$) and 5.04×10^{-7} ($\kappa = -11$).

IV. RESULTS: THRESHOLD-AMPLIFIED BREAKING OF SPIN SYMMETRY

Table II collects the production values for the $2p$ doublet ($\kappa = +1$: $2p_{1/2}$; $\kappa = -2$: $2p_{3/2}$) for Z from 1 to 130 at two reduced screenings, with grid spreads of 10^{-11}–10^{-10} throughout. Figure 1 summarizes the physics.

Near-threshold amplification. At $s = 0.24$ the width ratio $\Gamma_{2p_{3/2}}/\Gamma_{2p_{1/2}}$ grows monotonically: 1.000, 1.064, 1.333, 2.404, 77.5 for $Z = 1, 20, 40, 60, 80$. Beyond $Z \approx 92$ the ratio is formally infinite: the $2p_{1/2}$, whose critical screening exceeds 0.24 in exactly this range according to the bound-state map of Ref. [16], remains bound while its fine-structure partner is a broad resonance. The near-degeneracy of spin-doublet widths observed at high ℓ and weak effective coupling [13] is thus completely broken in the atomic strong-coupling regime.

A threshold effect. At $s = 0.30$, far above both thresholds, the same doublet gives ratios of at most 1.14 up to $Z = 60$. The amplification therefore does not reflect a uniformly growing relativistic width shift: it is controlled by the distance of each partner to *its own* critical screening, to which the width is exponentially sensitive. The splitting of the critical screenings $\lambda_c(Z, \kappa)$ — itself a purely relativistic, threshold-map effect — is transcribed into orders of magnitude in the widths near threshold. Consistently, the resonances are born at the critical screenings of the companion map: at $Z = 40$, $\kappa = -2$ the extrapolated birth point is $s_c = 0.2207$ against the bound-state value 0.216(4), and the width rises from threshold with the Wigner p-wave law, measured exponent $d \ln \Gamma / d \ln E_r = 1.530$ against the theoretical 3/2.

Sign inversion of the position splitting. Near threshold $E_r(2p_{3/2}) > E_r(2p_{1/2})$, while deep in the continuum the order reverses (at $Z = 40$, $s = 0.30$: 9.230 vs. 8.979). Splitting inversions between different doublets were noted in the nuclear-style cases of Ref. [13]; here a single doublet inverts between regimes.

V. CONCLUSIONS

Two independent, mutually blind methods — a certified complex-scaled exact-structure discretization and Gamow shooting matched to exact free spinors — agree to ten–twelve digits on Dirac–Yukawa resonances, turning the four-digit state of the art into benchmark tables, correcting its narrow strong-coupling entries, and resolving widths previously reported as zero. The physics outcome is a sharp statement about relativistic symmetry in autoionization: the spin symmetry of resonance widths, excellent at high angular momentum and away from thresholds, is broken by orders of magnitude near the critical screening, where the relativistic splitting of the thresholds themselves is transcribed exponentially into the widths — to the point that one fine-structure partner autoionizes while the other remains bound. The natural continuations are the d-doublets and the supercritical finite-nucleus regime, and, on the methods side, the finite-field problem where the same certified machinery meets non-separable perturbations.

TABLE II. Production values for the $2p$ doublet of $V = -Ze^{-\lambda r}/r$: E_r and Γ (a.u.), quoted at the grid-spread precision ($\leq 10^{-10}$ unless noted). "bound" marks a partner below its critical screening; "dissolved" a resonance too broad for the $\theta \leq 0.8$ window.

Z	s	$2p_{1/2}$ ($\kappa = +1$)		$2p_{3/2}$ ($\kappa = -2$)	
		E_r	Γ	E_r	Γ
1	0.24	0.002179006	0.002363910	0.002179167	0.002364262
1	0.30	0.005566663	0.018760420	0.005566558	0.018760950
20	0.24	0.840624043	0.878730	0.867171835	0.935150
20	0.30	2.247795246	7.406652	2.231161256	7.493011
40	0.24	2.941245536	2.711104	3.413196282	3.613002
40	0.30	9.229777734	28.37727	8.979248887	29.83296
60	0.24	4.598969039	3.181289	7.460504208	7.646312
60	0.30	21.50465234	58.52595	20.40804704	66.58487
80	0.24	0.848891758	0.159698	12.67315047	12.37604
80	0.30	39.11712762	87.92655	dissolved	
92	0.24	bound		16.13746549	15.16700
92	0.30	50.98258585	97.05267	dissolved	
110	0.24	bound		21.34539206	18.62827
110	0.30	60.90692212	77.12090	dissolved	
130	0.24	bound		26.14562678	20.34203
130	0.30	bound		dissolved	

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